LAYERSTONE

Action Plan & Implementation Roadmap

Step-by-step guide to building a 3D printed housing social enterprise

London + Birmingham | Solo Founder | Pilot-First

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1. Pre-Launch Checklist

Complete before spending any significant money. Estimated: 2–4 weeks.

1.1 Legal Structure Decision

- ☐ Decide Path A (CIC) or Path B (RHP) (review Section 7 of business plan)
- ☐ If Path A. Register CIC with Companies House (~£30 + £500 legal)
- ☐ If Path B. Begin RHP application to Regulator of Social Housing (6–12 month process)
- ☐ Open business bank account (Starling Business or Tide (free))
- ☐ Register for Corporation Tax with HMRC
- ☐ Get professional indemnity insurance (~£300–£500/year)

Cost: £500–£1,000 (Path A) or £5,000–£10,000 (Path B)

Timeline. Week 1–2

1.2 Technology Partner Outreach

- ☐ Research Icon's partnership/licensing models (email their business development team)
- ☐ Research COBOD's European partnership terms (they're geographically closer)
- ☐ Prepare a one-page partnership proposal (what Layerstone offers. UK regulatory expertise, land access, housing provider credentials)
- ☐ Identify any UK-based 3D concrete printing companies or university research groups
- ☐ Set meetings with 2+ potential technology partners within 60 days

Cost: £0 (your time only)

Timeline. Week 1–3

1.3 Council & Land Research

- ☐ Identify 5+ London boroughs with brownfield land and severe housing need
- ☐ Research each borough's Local Plan for housing targets and innovation appetite
- ☐ Shortlist 3 boroughs. Barking & Dagenham, Croydon, and Enfield recommended
- ☐ Begin informal contact with council housing departments (phone/email introducing the concept)
- ☐ Research Public Land for Housing Programme. Identify surplus government sites in target boroughs
- ☐ Research Brownfield Land Release Fund (identify eligible sites)
- ☐ Begin Birmingham research in parallel (Birmingham City Council regeneration team)

Cost: £0–£100 (travel)

Timeline. Week 2–4

1.4 Brand & Digital Presence

- ☐ Secure domain name (layerstone.co.uk) (~£10/year)
- ☐ Create landing page (Carrd.co £19/year or free WordPress)
- ☐ Set up professional email (Google Workspace £5/month)
- ☐ Create LinkedIn company page
- ☐ Design logo (Canva free or Fiverr £20–£50)
- ☐ Write a one-page vision document for council and partner conversations

Cost: £50–£150

2. Phase 1: Foundation & Partnerships (Months 1–3)

Goal. Secure a technology partner commitment and at least one council relationship.

2.1 Technology Partnership

- Hold initial meetings with Icon and/or COBOD business development
- Negotiate terms. Licensing, equipment lease, joint venture, or printing-as-a-service
- Agree pilot scope (a 2–5 unit demonstration project)
- If no international partner responds. Explore UK university partnerships (Loughborough, Nottingham (both have 3DCP research labs))
- Begin BBA (British Board of Agrément) certification process for the concrete mix (~£5,000–£10,000, takes 3–6 months)

2.2 Council Engagement

- Formal introductions to housing departments in 3 target boroughs
- Present the Layerstone concept: 3D printed homes, 40–60% cost saving, income-proportional rent
- Ask about available brownfield or council-owned sites
- Offer to present to council housing committees
- Identify a council 'champion'. One officer who sees the innovation value

2.3 Funding Applications

- Apply to Community Housing Fund (if Path A CIC)
- Apply to Affordable Homes Programme (if Path B RHP (may need to wait for registration))
- Research social investment opportunities. Big Society Capital, Social Investment Business
- Prepare investor deck and financial model for impact investors

2.4 Regulatory Preparation

- Engage a structural engineer familiar with 3DCP or innovative construction
- Commission preliminary structural testing of proposed concrete mix
- Contact BBA to begin certification process
- Identify Building Control body for pilot area (either council BC or approved inspector)

2.5 End-of-Phase Checkpoint

☐ Do I have a technology partner committed (or in advanced discussions)?

☐ Is at least one council interested in a pilot site?

☐ Is BBA certification in progress?

☐ Do I have a funding plan for the first build?

If 3/4 YES → proceed to Phase 2. If not → extend by 1–2 months.

3. Phase 2: Site Acquisition & Planning (Months 4–6)

Goal. Secure a pilot site and submit planning application.

3.1 Site Selection

- Shortlist 2–3 specific sites from council discussions
- Commission site surveys (ground conditions, contamination, access)
- Negotiate land terms. Lease, purchase at below-market rate, or Public Land for Housing allocation
- Preferred. Brownfield site with existing access and no conservation constraints

Criterion	Weight	Site A	Site B	Site C
Cost (subsidised)	25%	Score 1-10	Score 1-10	Score 1-10
Planning likelihood	25%			
Transport links	15%			
Housing need in area	15%			
Community sentiment	10%			
Ground conditions	10%			

3.2 Planning Application

- Submit pre-application to Local Planning Authority (LPA) (£500–£2,000 fee)
- Attend pre-application meeting with planning officers
- Commission architect drawings (can be digital-native from 3D printer design files)
- Prepare Design & Access Statement
- Begin community consultation (leaflets, local meetings)
- Submit full planning application (target 8–13 weeks determination)

3.3 Funding Secured

- Confirm Community Housing Fund grant (if applied)
- Secure social investment commitment (Big Society Capital or equivalent)
- Have bridge funding in place for gap between land acquisition and first rental income

3.4 Birmingham Parallel Track

- Identify 2–3 potential sites in Birmingham
- Begin informal council discussions
- This runs at 50% intensity (London is the priority)

3.5 End-of-Phase Checkpoint

☐ Is a site secured or under agreement?

☐ Has planning pre-application received positive signals?

- ☐ Is funding committed for the first build?
- ☐ Is BBA certification progressing?

If 3/4 YES → proceed to first build. If planning is refused → appeal or find alternative site.

4. Phase 3: First Build (Months 7–9)

Goal. Print and complete 2–3 homes. Get Building Control sign-off.

4.1 Construction

- Planning permission granted (or commenced under permitted development if applicable)
- Technology partner deploys 3D printer to site
- Print structural shells for first 2–3 units (12–24 hours each)
- Install foundations (conventional poured concrete slab (done before printing))
- Fit-out. Roof structure, insulation (Part L compliance), windows, doors
- Services. Plumbing, heating, electrics (qualified trades)
- Internal finishing. Kitchen, bathroom, plastering, painting
- External. Landscaping, parking, pathways

4.2 Compliance & Sign-Off

- Building Control inspections at each stage (foundation, structure, services, completion)
- Provide BBA certification documents and structural test results
- EPC (Energy Performance Certificate) assessment
- Building Control completion certificate issued
- Arrange structural warranty (NHBC or bespoke if NHBC won't cover 3DCP)

4.3 Cost Tracking

Item	Budget	Actual	Variance
Land	£40,000/unit	TBC	
3D printing	£20,000/unit	TBC	
Fit-out	£47,000/unit	TBC	
Compliance	£8,000/unit	TBC	
Contingency (15%)	£17,000/unit	TBC	
TOTAL	~£130,000/unit	TBC	

Track actual costs meticulously. This data is the case study that sells every future project.

5. Phase 4: First Tenants & Validation (Months 10–12)

Goal. Move tenants in, collect first rent, publish case study.

5.1 Tenant Selection

- Partner with council housing team for referrals (social rent tenants)
- Advertise affordable rent units through housing portals and council lists
- Income verification process for proportional rent calculation
- Tenancy agreements drafted (AST. Assured Shorthold Tenancy, or if RHP. Secure tenancy)
- Move-in support. Appliance packs, tenant handbook, welcome visit

5.2 Property Management

- Set up maintenance reporting system (simple. WhatsApp group + email)
- Arrange contents insurance offering for tenants
- Schedule 3-month post-completion inspection
- Begin collecting rent and Housing Benefit payments

5.3 Case Study & PR

- Document the full build process. Costs, timeline, challenges, solutions
- Photograph finished homes and record tenant testimonials (with permission)
- Publish case study on website and LinkedIn
- Approach trade press. Inside Housing, Construction News, Property Week
- Present to other councils and housing associations

5.4 End-of-Year Checkpoint

- ☐ Are all pilot units occupied with paying tenants?
- ☐ Did actual costs come within 15% of budget?
- ☐ Did Building Control sign off without major issues?
- ☐ Is there demand for more units (pipeline of interested tenants)?
- ☐ Is the technology partner satisfied and willing to continue?

If 4/5 YES → expand. If costs were wildly over budget → investigate before building more.

6. Phase 5: Expansion (Months 13–18)

Goal. Build remaining pilot units, secure Birmingham site, begin Phase 2 social programme design.

6.1 London Expansion

- Build remaining pilot units (total 5–10 in first development)
- Begin second London site identification
- Negotiate terms for larger development (15–20 units)
- If Path A. Begin RHP registration process with track record from pilot

6.2 Birmingham Launch

- Secure first Birmingham site (brownfield or council-owned)
- Submit planning application
- Deploy printer for first Birmingham build (2–3 units)
- Partner with Birmingham council housing team for tenant referrals

6.3 Team Growth

Trigger	Hire	Cost	Funded By
5+ units under management	Part-time property manager	£20K–£25K	Rental income
Second site in planning	Project coordinator	£28K–£32K	Grant income
If Path A → RHP transition	Compliance officer (part-time)	£15K–£20K	Operating budget
Birmingham site active	Birmingham site manager	£25K–£30K	Birmingham funding

6.4 Phase 2 Social Programme Design

- Research homelessness charity partners (Shelter, Crisis, St Mungo's, local organisations)
- Design employment training curriculum: 3D printer operation, property maintenance
- Draft homeless rehousing policy. Referral process, rent-free period, transition plan
- Apply for social programme funding (separately from housing construction funding)

7. Phase 6: Scale (Year 2–3)

Goal: 20–50 units across London and Birmingham. Establish Layerstone as a recognised housing provider.

7.1 Portfolio Growth

- Target: 20 units by end of Year 2, 50 units by end of Year 3
- Develop on multiple sites simultaneously (2–3 active sites)
- Standardise the build process. Repeatable playbook for each site
- Build relationships with housing associations for partnership delivery

7.2 Funding Maturity

- If Path A. Complete RHP registration. Unlocks Affordable Homes Programme grants
- Access Housing Finance Corporation loans at preferential rates (2–3%)
- Consider community bond issue (raise £100K–£500K from impact investors)
- Cross-subsidise. Profitable units fund social rent units

7.3 Phase 2 Social Programme Launch

- First homeless rehousing tenants (10–20% of new units allocated)
- First cohort of employment training programme
- Publish social impact report
- Media coverage. Homelessness angle attracts significant press interest

8. Decision Gates & Kill Criteria

Non-negotiable checkpoints. Be honest with yourself at each gate.

Gate	Timing	Continue If...	Pivot If...	Kill If...
Gate 1	Month 3	Tech partner engaged, 1+ council interested	No partner but council interest	No partner AND no council interest
Gate 2	Month 6	Site secured, planning positive	Planning difficult but solvable	Planning refused on all sites, no alternatives
Gate 3	Month 9	First homes printed, costs on target	Costs 15–30% over budget	Costs 50%+ over budget, tech not working
Gate 4	Month 12	Tenants in, rent collecting, case study done	Slow uptake but demand exists	No tenants, no demand, costs unsustainable
Gate 5	Month 18	10+ units, Birmingham secured	Growth slower than planned	Model not replicable, one-off success only

8.1 Pivot Options

- If 3D printing fails regulatory hurdles → pivot to modular construction (same land, same social model)
- If land costs are prohibitive → partner with existing housing association (they provide land, you provide construction)
- If funding is insufficient → consulting model (advise councils/HAs on 3DCP, earn fees while building pipeline)
- If London is too expensive → start in Birmingham or another secondary city

Every pivot still delivers affordable housing. The 3D printing technology is a means, not the mission.

9. Budget Timeline

9.1 Pre-Launch & Phase 1 (Months 1–3)

Item	Cost (Path A)	Cost (Path B)	Essential?
Legal setup	£500–£1,000	£5,000–£10,000	Yes
Insurance	£300–£500	£500–£1,000	Yes
Domain + email + branding	£100–£200	£200–£500	Yes
Travel (council/partner meetings)	£300–£500	£500–£1,000	Yes
BBA certification (deposit)	£2,500–£5,000	£2,500–£5,000	Yes
Structural testing	£1,500–£3,000	£1,500–£3,000	Yes
TOTAL Phase 1	£5,200–£10,200	£10,200–£20,500	

9.2 Phase 2–3: Site & Build (Months 4–12)

Item	Cost	Funded By
Land acquisition (subsidised, 5 units)	£150,000–£250,000	Grants + social investment
3D printing (5 units)	£75,000–£125,000	Grants
Fit-out (5 units)	£150,000–£225,000	Grants + investment
Compliance & fees	£25,000–£40,000	Operating budget
Operating costs (12 months)	£15,000–£25,000	Savings + early revenue
TOTAL Year 1 Build	£415,000–£665,000	

You do not need £665K upfront. Construction costs are staged. Land first, then print, then fit-out. Each stage can be funded in tranches as grants and investment arrive.

10. Risk Contingencies

Risk	Contingency	Cost of Contingency
Planning refused	Appeal (£5K–£15K) or move to alternative site	2–4 months delay
Tech partner withdraws	Engage backup partner (COBOD if Icon fails) or UK university	1–3 months delay
Building Control rejects 3DCP	Additional testing + BBA certification. Worst case: build traditionally on same site	£5K–£15K + 3 months
Construction cost overrun	15% contingency budgeted per unit. If >30% over: pause and reassess	Budgeted
Grant application rejected	Reapply next round + pursue alternative funding (social bonds, angel investors)	3–6 months delay
Council relationship breaks down	Move to alternative borough — 3 target boroughs prevents single-point failure	1–2 months
Cash flow gap (build cost before rent)	Bridge finance from social lender. Invoice grant tranches against milestones	Interest cost
Founder burnout	Hire property manager at 5-unit trigger. Take planned holidays. This is a marathon, not a sprint	£20K–£25K

11. Key Milestones Calendar

Month	Milestone	Success Metric	Decision
Month 1	Legal entity registered	CIC/RHP application submitted	—
Month 2	Tech partner meeting held	LOI or partnership terms discussed	—
Month 3	GATE 1	Partner engaged + council interested	Continue / Pivot
Month 4	Site shortlisted	2+ viable sites identified	—
Month 5	Pre-application submitted	LPA pre-app meeting attended	—
Month 6	GATE 2	Site secured + planning positive	Continue / Pivot
Month 7	Planning submitted	Full application lodged	—
Month 8	First home printed	Structural shell complete	—
Month 9	GATE 3	2–3 homes printed + costs on target	Continue / Pivot
Month 10	First home completed	Building Control sign-off	—
Month 11	First tenant moves in	Rent collection begins	—
Month 12	GATE 4	All pilot units occupied + case study	Continue / Pivot
Month 15	Birmingham site secured	Site + planning in progress	—
Month 18	GATE 5	10+ units, model replicable	Scale / Consolidate
Month 24	Year 2 complete	20+ units, revenue covering costs	—
Month 36	Year 3 complete	50+ units, RHP registered (if Path A)	—

Final Notes

This action plan is designed for a solo founder building something that has never been done in the UK. A 3D printed affordable housing social enterprise.

The plan is deliberately cautious. Every phase has a gate. Every gate has a pivot option. The worst-case scenario is not failure. It's a pivot to conventional construction with the same social mission and land relationships intact.

The technology is proven internationally. The regulatory framework exists but hasn't been tested with 3DCP. You will be the pioneer. That means slower timelines, more conversations with Building Control, and more documentation than a traditional builder would face.

But it also means first-mover advantage in a market that every housing association, every council, and every government minister knows is broken.

Build layer by layer. Prove the model. Scale what works.

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